Analysis of Factors Affecting Students' Daily Study Hours Using Heteroskedasticity-Robust Regression Models

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***Abstract* — Students' daily study hours are influenced by academic, behavioral, and lifestyle factors, yet survey-based analyses are often affected by heteroskedasticity, leading to unreliable inference under conventional Ordinary Least Squares (OLS). This study examines the determinants of undergraduate students' study hours outside of class using a multivariate regression framework with heteroskedasticity-robust estimation.**

Using survey data from 45 undergraduate students, eight explanatory variables are analyzed, including academic workload, sleep duration, work hours, organizational activities, and social media usage. Diagnostic testing confirms the presence of heteroskedasticity. Comparative analysis shows that OLS with HC3 robust standard errors provides the most reliable and interpretable inference. Across all robust specifications, organizational activity hours consistently emerge as the strongest predictor of study time, while sleep duration and work hours exhibit estimator-sensitive effects.

The results emphasize the importance of robust regression methods in survey-based educational research.

Keywords* — *Study Hours; Heteroskedasticity; Robust Regression; Ordinary Least Squares; Student Time Allocation

I. INTRODUCTION

In recent years, university students in Indonesia have faced increasing tension between academic demands and lifestyle changes driven by digital technology, part-time work, and campus organizational activities. National statistics report that Indonesian students studied outside school hours for an average of 70.16 minutes per day in 2024, with higher-education students studying independently for approximately 83.26 minutes per day [1]. This study time must compete with online engagement, social media usage, commuting, work commitments, and extracurricular involvement, all of which may either support or reduce students' effective daily study hours [2].

Previous research highlights the importance of sleep quality as a determinant of learning concentration and academic readiness. An Indonesian study on high-school learners found that better sleep quality significantly improved learning concentration ($\beta = 0.154$; $p = 0.005$), with intrinsic motivation strengthening this effect [3][4]. These findings suggest that students with healthier sleep patterns are better able to sustain attention during study sessions, potentially increasing the productivity of their daily study hours.

Digital behavior has also become a major factor influencing academic outcomes. A large-scale study involving 3,716 Chinese undergraduates reported that high-frequency social media use had a significant negative effect on academic performance, partly mediated by social anxiety and fear of missing out [5]. Although conducted in a different context, these behavioral mechanisms mirror trends among Indonesian students given the country's high internet penetration and widespread social media usage [6]. Poorly managed online activity can diminish both the quantity and quality of study time.

Non-academic commitments such as part-time work and organizational involvement further contribute to variations in study behavior. Evidence suggests that part-time employment may negatively affect academic performance when it leads to fatigue or time pressure, whereas campus organizational participation may enhance academic outcomes by developing soft skills, discipline, and time-management habits [7][8]. These findings indicate that not all time away from formal studying is detrimental; some activities can indirectly benefit academic routines when balanced effectively.

Physical and logistical factors also play an important role. A systematic review of nineteen intervention studies found that physical activity commonly supports cognitive functions such as working memory and executive functioning, producing generally positive or neutral academic effects [9]. In contrast, empirical evidence shows that longer commuting durations such as an additional ten minutes of travel are associated with lower standardized test performance, suggesting that extended travel time reduces available study hours and increases fatigue [10].

Given these diverse influences, it becomes essential to examine how variables such as age, number of credits (SKS), sleep duration, commute time, working hours, organizational involvement, physical exercise, and social media use jointly determine students' daily study duration. At the same time, the statistical model used to analyze these relationships must be reliable. Heteroskedasticity, non-

constant variance in regression errors is frequently observed in cross-sectional education data and can bias standard errors and hypothesis testing [11][12][13]. Therefore, this study not only identifies the most influential predictors but also compares several heteroskedasticity-robust regression methods, including Weighted Least Squares (WLS), robust standard errors, Robust WLS (RWLS), and Generalized Least Squares (GLS), to determine the most appropriate model for analyzing students' daily study hours.

Research Questions

1. How does the use of heteroskedasticity robust regression models improve the accuracy of estimating students' daily study hours?
2. Which variable predictor has the strongest influence on study hours?
3. How do these eight variables affect daily study hours in a multivariate regression framework?

Objectives

1. To evaluate the performance of OLS, WLS, robust standard errors, RWLS, and GLS in modeling students' daily study hours.
2. To identify the most influential predictor among the eight variables.
3. To analyze both partial and simultaneous effects of the independent variables on students' daily study duration.

Scope

1. The study focuses on undergraduate students' self-reported daily study hours and eight predictors: age, SKS load, sleep duration, commuting time, work hours, organization involvement, exercise hours, and social media usage.
2. The analysis is limited to regression-based techniques addressing heteroskedasticity: OLS, WLS, robust SE, RWLS, and GLS.
3. External psychological constructs such as motivation, personality, and institutional specific variables are outside the scope.
4. The dataset reflects a specific student population and findings that may not generalize beyond this demographic.

Research Hypothesis

1. We believe that heteroskedasticity-robust regression methods including WLS, robust standard errors, Robust WLS (RWLS), and GLS will produce more reliable estimates than standard OLS when modeling students' daily study hours. This expectation reflects evidence that cross-sectional education data often exhibit non-constant error variance, making robust variance estimators more appropriate for inference.
2. We believe that all eight predictors included in the model; age, SKS load, sleep duration, commuting time, part-time work hours, organizational involvement, exercise hours, and social media usage contribute meaningfully to explaining variations in students' daily study hours. Each variable represents a distinct aspect of students' behavioral, academic, or lifestyle conditions that has been shown in prior research to influence learning routines.
3. We believe that certain predictors will show consistent directional effects, particularly that sleep duration will have a positive influence on study hours, while commuting time, part-time work hours, and social media usage are expected to reduce effective study duration. Conversely, organizational activity and physical exercise are hypothesized to have neutral or positive effects. These expected relationships provide a theoretical basis for evaluating the significance and strength of each predictor within heteroskedasticity-robust regression models.

Research Gap

Existing literature mainly examines single factors such as sleep quality [3], motivation [4], commuting time [10], part-time work [7], or social media usage [5] without integrating them into a unified multivariate framework. Few studies apply heteroskedasticity-robust regression techniques despite documented issues of inconsistent variance in education-related survey data [11][12][13]. Current studies also tend to focus on academic performance rather than daily study duration, leaving a gap in understanding the behavioral structure underlying study time itself [14][15][16]. This research addresses these limitations by examining eight predictors simultaneously while comparing OLS, WLS, robust SE, RWLS, and GLS to determine the most reliable modeling approach.

II. THEORITICAL REVIEW

A. Students' Study Hours and Behavioral Factors

Daily study hours describe how much time students spend learning on their own outside scheduled classes. National survey data show that Indonesian university students average 83.26 minutes/day of study time [1]. Sleep quality improves attention and learning concentration [5], strengthened by intrinsic motivation [4]. Social media usage, particularly high-frequency exposure, reduces academic focus [5]. Commuting time reduces effective learning time due to fatigue and time displacement [10]. Part-time work may reduce academic engagement if work hours are high [7]. Organizational involvement can improve discipline and time management [8], while physical exercise benefits cognitive performance [9]. These findings justify modeling multiple behavioral predictors simultaneously.

B. Multiple Linear Regression (OLS)

Multiple Linear Regression (Ordinary Least Squares or OLS) is a statistical approach used to model the linear relationship between one continuous variable responds to several predictors [11]. The general form of the OLS model is:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_k X_k + \epsilon$$

where Y is the response variable, X_1 to X_k are the predictor variables, β_0 is the intercept, the expected value

of Y when all predictors are zero, β_j (for j = 1…k) are the regression coefficients, which show how much Y changes for a one-unit increase in each predictor, and ϵ is the error term.

OLS assumes linearity, independence, normality of residuals, and homoskedasticity (constant variance of errors). When these assumptions hold, OLS produces the Best Linear Unbiased Estimators (BLUE) [11]. However, violations particularly heteroskedasticity, which is common in survey-based education datasets can lead to biased standard errors and unreliable inference.

C. Heteroskedasticity

Heteroskedasticity occurs when the variance of errors is not constant. In simple terms, the model errors get bigger or smaller depending on the level of certain predictors. Even though heteroskedasticity does not change the OLS coefficient estimates, meaning the estimated β's remains unbiased, it does affect how confident we can be about those estimates. The biggest problem appears in the standard errors [13].

When heteroskedasticity is present, OLS underestimates or overestimates the true variability of the coefficients; standard errors become incorrect, that leads to wrong t-tests, incorrect p-values, and misleading confidence intervals. A predictor might appear significant even when it is not or looks insignificant when it actually matters. Because of these issues, analysts often use methods that adjust for or directly model the unequal variance [17] [18].

D. Weighted Least Squares (WLS)

WLS tries to fix the problem at its source by modifying how the regression is estimated [12]. Instead of assuming equal variance, WLS gives each observation a weight that is inversely proportional to its error variance. Observations with large variance, given smaller weight. Observations with small variance, given larger weight. The estimator becomes:

$$\beta_{WLS} = (X'WX)^{-1} X'WY$$

Where W is a diagonal matrix containing weights

$$w_i = \frac{1}{\sigma_i^2}.$$

Because the true variances are unknown, they are estimated from an initial OLS model (usually using the squared residuals). Then the weights are computed, and the model is re-estimated using WLS. This produces more efficient and more reliable estimates compared to OLS. Prior empirical studies show that WLS can outperform transformation-based methods in correcting variance instability [12].

E. Robust Standard Errors (HC3)

Robust Standard Errors especially the HC3 version introduced by White [13] adjusts the variance–covariance matrix of OLS estimators to remain valid under heteroskedasticity without altering coefficient estimates. HC3 is widely used in econometric applications. HC3 works by giving additional corrections to observations with large leverage, which makes it more conservative and more stable for small or moderate sample sizes [19].

F. Robust Weighted Least Squares (RWLS)

RWLS integrates Weights are added (like WLS) to handle unequal error variance, and Robust standard-error adjustments are applied afterward. This approach becomes helpful when heteroskedasticity is accompanied by influential points or moderate outliers. RWLS stabilizes both the coefficient estimates and the standard errors, giving more trustworthy inference [11][12].

G. Generalized Least Squares (GLS)

GLS goes one step further by trying to model the structure of the error variance directly. If the form of heteroskedasticity can be approximated. For example, following a power-variance or proportional-variance pattern, GLS transforms the model so that the errors become homoskedastic in the transformed space [20]. Its estimators are obtained by:

$$\beta_{GLS} = (X'\Sigma^{-1}X)^{\{-1}X'\Sigma^{-1}Y$$

where $\boldsymbol{\Sigma}$ is the variance–covariance matrix of the errors. When the structure is correctly specified, GLS is highly efficient and produces smaller standard errors than OLS.

H. Literature on Factors Affecting Study Hours

Prior findings consistently show that sleep supports academic concentration [3][4], commuting reduces study time [10], part-time work can compromise study duration [6], organizational involvement may foster discipline [8], physical exercise benefits cognition [9], and social media distractions reduce academic focus [5]. These findings support the inclusion of all eight predictors in the regression models.

III. Research Methodology

A. *Data Collection & Data Preparation*

1. Dataset Source
   The dataset was collected through an online survey of undergraduate students aged 18–23 years old. Respondents reported their daily routines including study hours, sleep duration, social media usage, exercise, commuting time, part-time work, and organizational activities. The survey also included non-time variables such as age and Semester Credit Units (SKS). The dataset was specifically designed to analyze factors that influence students' daily study hours.

2. Data Cleaning
   Data cleaning was conducted as a preliminary step to ensure that the dataset was appropriate for

subsequent analysis. The dataset was restricted to respondents with a *Mahasiswa* status to maintain a consistent and relevant study population aligned with the research objectives. Variables containing personal identifiers were removed to preserve respondent anonymity and to ensure that the analysis focused solely on academic and behavioral attributes. This step aimed to improve data consistency and reduce potential noise arising from irrelevant or heterogeneous observations.

3. Handling Missing Values and Outliers
The dataset was examined for missing and inconsistent values as part of quality assurance. This process ensured that the data used for analysis were complete and logically valid. Implausible entries that violated realistic constraints were identified and addressed to prevent distortion of subsequent statistical modeling. In contrast, extreme values that remained within reasonable bounds were retained, as they may reflect genuine variation in students' daily behaviors. Overall, this step aimed to enhance the reliability of the dataset without unnecessarily altering its natural variability. Specific implementation details and final exclusion decisions are described in the Data Preprocessing subsection.

B. *Exploratory Data Analysis (EDA)*

Exploratory Data Analysis (EDA) was employed to gain an initial understanding of the distributional characteristics of the variables and their potential relationships with daily study duration (*StudyHoursPerDay*). Through graphical and summary-based techniques, EDA serves to reveal patterns, assess variability, and identify potential issues that may influence modeling assumptions. Rather than drawing inferential conclusions, this stage provides insight into the structure of the data and informs the choice of appropriate statistical methods for subsequent regression analysis [21].

C. *Model Selection and Justification*

Since the dependent variable, StudyHoursPerDay, is continuous, Multiple Linear Regression using Ordinary Least Squares (OLS) was selected as the baseline model [22]. However, because the data are cross-sectional and derived from self-reported surveys, heteroskedasticity is likely to be present. To ensure reliable statistical inference, heteroskedasticity-robust regression methods including Weighted Least Squares (WLS), robust standard errors (HC3), Robust WLS (RWLS), and Generalized Least Squares (GLS) [23][24] were employed and compared with the OLS model.

To complement the theory-driven full specification, variable selection procedures were conducted to identify a parsimonious alternative model. Forward selection and backward selection were applied based on statistical significance, with predictors added or removed according to their p-values [25]. These procedures were not intended to replace the theory-driven full model, but rather to serve as a robustness check and to provide a parsimonious benchmark specification for comparison purposes. The results of the variable selection process are presented and discussed in the subsequent analysis section.

D. *Data Preprocessing*

The dataset was filtered to include only respondents with student status (*Mahasiswa*). Identifying variables were removed to preserve anonymity, and all analytical variables were converted to numeric format. Prior to model estimation, the data were screened for implausible entries and influential observations. Observations containing clearly invalid values (e.g., unrealistic daily activity duration or incorrect academic credit input) and exhibiting excessive influence were excluded. The final dataset used for analysis consisted of 45 undergraduate students.

E. *Model Evaluation*

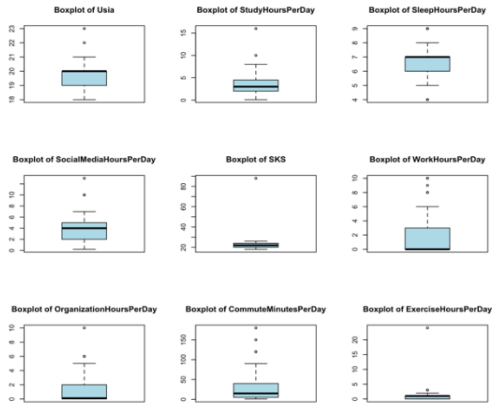
Model performance was evaluated using overall model significance, goodness-of-fit measures, and inferential stability under heteroskedasticity. Comparisons between OLS and heteroskedasticity-robust estimators focused on coefficient consistency and standard error reliability rather than changes in point estimates alone.

IV. DATA ANALYSIS

A. *Exploratory Data Analysis*

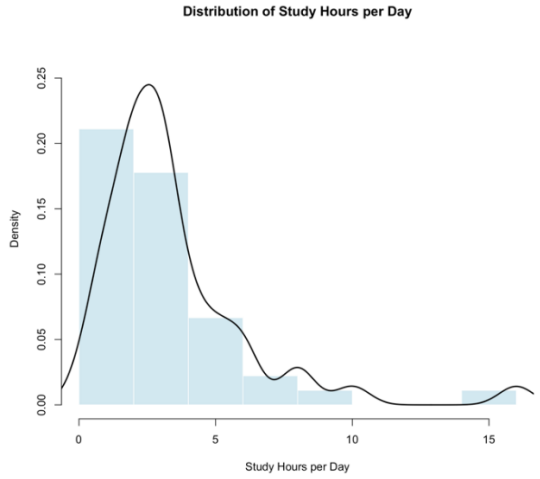
1. Boxplot Distribution of Each Variables

The boxplot below presents boxplots of the main numerical variables to provide a concise overview of their distributions and to identify potential outliers. Overall, most variables exhibit moderate dispersion with a small number of observations lying beyond the whiskers. Variables such as Study Hours Per Day, Social Media Hours Per Day, Work Hours Per Day, Organization Hours Per Day, and Commute Minutes Per Day show slight right-skewness, indicating that while most students report relatively typical daily activity durations, a few respondents experience heavier workloads or longer activity times. These extreme values remain within a plausible range and may reflect genuine variations in students' daily routines rather than data errors.



In contrast, the boxplots reveal clearly implausible extreme values in two variables. The SKS variable contains an observation far exceeding the normal academic credit load, which is inconsistent with standard university regulations and thus unlikely to represent a valid response. Similarly, ExerciseHoursPerDay shows an extreme value corresponding to nearly 24 hours of exercise per day, which is not realistically feasible. These observations are considered unrealistic rather than merely extreme. Consequently, while most detected outliers are treated as reasonable variations, the extreme values in SKS and ExerciseHoursPerDay warrant special attention and are addressed separately in the data preparation stage prior to regression analysis.

2. Histogram



The histogram, combined with the overlaid density curve, illustrates the distribution of study hours per day among students. The distribution is right-skewed, with most observations concentrated in the low to moderate range of study hours, indicating that the majority of students spend a similar amount of time studying each day. A small number of observations form a long right tail, reflecting students who report substantially higher study durations. Overall, the distribution does not show extreme irregularities after data cleaning and provides a reasonable basis for subsequent regression analysis.

B. Data Preparation Analysis

1. Data Cleaning
   Prior to analysis, the dataset was filtered to include only respondents with a student status (*mahasiswa*) to ensure relevance to the research objectives and sample consistency.

TABEL I. STATUS RESPONDENTS

Status	Frequency
Mahasiswa	47
Married	1

As shown in the table above, the dataset is dominated by student respondents. The single non-student observation was excluded from further analysis to maintain a homogeneous study population.

2. Handling missing value
   A missing value inspection was conducted after data cleaning. The results indicate that no missing (NA) values are present in the dataset. Therefore, no imputation or removal of observations was required, and the dataset was considered complete for subsequent analysis.

3. Outlier Handling
   Outlier detection was conducted as part of data quality assessment to identify observations that could potentially distort the analysis. The Interquartile Range (IQR) method was used as a reference to flag extreme values across numerical variables. Most detected outliers were found to be within reasonable bounds and were therefore retained, as they may represent genuine variation in students' daily activities.

TABLE II. OUTLIER DETECTIONS

Variable	Number of Outliers
SKS	1
ExerciseHoursPerDay	3

From the outliers identified, only extreme and logically implausible observations were removed. Specifically, the single extreme value in SKS, which exceeded reasonable academic credit limits, and one extreme value in ExerciseHoursPerDay, corresponding to an unrealistically high daily exercise duration, were excluded from the dataset. Other values flagged by the IQR method were retained, as they remained within plausible bounds and may represent genuine variation in students' daily activities.

TABLE III. SAMPLE SELECTION AND DATA CLEANING PROCESS

Step	Description	Remaining Observations
Raw data	Initial survey responses	48
Status filtering	Remove non-student respondents	47
Data validation	Remove implausible & influential observations (Exercise = 24h; SKS = 88)	45

C. Assumption Checking

Prior to formal statistical inference, the classical assumptions of multiple linear regression were evaluated using a hypothesis testing framework at a significance level of $\alpha = 0.05$.

1. Normality of Residuals
   Residual normality was assessed using the Liliefors (Kolmogorov-Smirnov) normality test.
   - H_0: The regression residuals are normally distributed.

- H_1: The regression residuals are not normally distributed.

The test yielded a p-value of 0.1884, which exceeds the significance level ($\alpha = 0.05$). Therefore, H_0 is not rejected, indicating that the residuals do not significantly deviate from normality. This result supports the normality assumption of the regression model.

2. Autocorrelation of Residuals

Serial correlation in the residuals was examined using the Durbin–Watson test.

- H_0: There is no positive autocorrelation in the regression residuals.
- H_1: There is positive autocorrelation in the regression residuals.

The Durbin–Watson statistic was DW = 1.87, with a corresponding p-value of 0.3119. Since the p-value exceeds $\alpha = 0.05$, H_0 is not rejected, indicating no evidence of positive autocorrelation. Although autocorrelation is more commonly associated with time-series data, this diagnostic is reported for completeness.

3. Homoskedasticity

The assumption of constant error variance was tested using the studentized Breusch–Pagan test.

- H_0: The regression residuals exhibit homoskedasticity (constant variance).
- H_1: The regression residuals exhibit heteroskedasticity (non-constant variance).

The test statistic was BP = 16.438, with a p-value of 0.0365, which is less than $\alpha = 0.05$. Thus, H_0 is rejected, providing statistical evidence of heteroskedasticity in the cross-sectional survey data. This finding justifies the use of heteroskedasticity-robust estimation methods, including robust standard errors, WLS/RWLS, and GLS, to obtain reliable inference.

4. Multicolinearity

Multicollinearity among the independent variables was assessed using the Variance Inflation Factor (VIF), a diagnostic measure that quantifies the extent to which the variance of a regression coefficient is inflated due to linear dependence among predictors.

Following commonly accepted guidelines, VIF values greater than 10 indicate severe multicollinearity, while values below 5 suggest no practical multicollinearity concern. The computed VIF values for all predictors ranged from approximately 1.14 to 1.39, which are well below these thresholds. This indicates that multicollinearity is negligible in the model, and the estimated regression coefficients are not adversely affected by linear dependence among the predictors.

Overall, the diagnostic tests indicate that the regression model satisfies the assumptions of residual normality, independence, and low multicollinearity. However, the presence of heteroskedasticity motivates the adoption of heteroskedasticity-robust regression techniques in subsequent analysis.

D. Variable Selection

To complement the theory-driven full specification, variable selection procedures were conducted to identify a parsimonious alternative model. Forward selection was implemented by first fitting single-predictor regressions and selecting the predictor with the smallest p-value for inclusion. OrganizationHoursPerDay showed the strongest univariate association with StudyHoursPerDay and was selected first. In the next step, candidate two-predictor models were evaluated by pairing OrganizationHoursPerDay with each remaining predictor. The addition of SleepHoursPerDay produced the most meaningful improvement and remained statistically significant, resulting in a two-predictor model including OrganizationHoursPerDay and SleepHoursPerDay.

Backward elimination was also performed starting from the eight-predictor model and iteratively removing the predictor with the largest p-value. This process converged to the same reduced specification containing OrganizationHoursPerDay and SleepHoursPerDay, providing consistent evidence that these variables are the most stable predictors in this dataset. Importantly, the reduced model is presented as a parsimonious benchmark, whereas the full model is retained as the primary specification due to theoretical motivation and to mitigate omitted-variable bias. The reduced model is therefore reported for comparison purposes (Appendix), while robust estimation is applied in subsequent analysis given the presence of heteroskedasticity.

E. Model Selection

While the forward and backward selection procedure produces a reduced model with stronger statistical significance and slightly lower complexity, the full eight variable model is maintained as the primary specification. The inclusion of all variables is theoretically motivated and prevents potential omitted variable bias. Moreover, the explanatory power of the full model remains high, as indicated by a relatively stable adjusted R^2. 0

1. Ordinary Least Squares (OLS)

A. Baseline Model OLS

The analysis begins with the estimation of a multiple linear regression model using Ordinary Least Squares (OLS). This baseline model serves as a benchmark for understanding the relationship between students' daily study hours outside of class and a set of demographics, academic, and activity-related factors.

TABEL IV. BASELINE OLS RESULTS

Variable	Coefficient	Std. Error	tStatistic	pValue
Intercept	13.8159	7.6905	1.796	0.0806
Age (Usia)	-0.3838	0.3420	-1.122	0.2690

Sleep Hours per Day	-0.5624	0.2489	-2.260	0.0298
Social media Hours per Day	-0.0920	0.1189	-0.774	0.4441
Academic Credit Load (SKS)	-0.0299	0.1368	-0.219	0.8282
Work Hours per Day	0.2549	0.1014	2.515	0.0164
Organization Hours per Day	0.8864	0.1453	6.103	0.0000
Commute Minutes per Day	0.0077	0.0069	1.113	0.2729
Exercise Hours per Day	0.0191	0.2401	0.0800	0.9367

Diagnostic testing on the baseline OLS model indicates the presence of heteroskedasticity, as evidenced by the studentized Breusch–Pagan test. Under heteroskedasticity, the conventional OLS standard errors are inconsistent, which may lead to unreliable statistical inference. So, the baseline OLS model cannot be used for final inference. To address this issue while preserving the original coefficient estimates, the model is re-estimated using heteroskedasticity robust standard errors (HC3) shown at part b below.

B. Robust OLS Estimation (HC3)

The HC3 estimator does not alter the OLS coefficient estimates. Instead, it adjusts the standard errors to remain valid in the presence of non-constant error variance. As a result, hypothesis testing based on robust t-statistics and p-values is consistent even when heteroskedasticity is present.

It is important to note that the application of HC3 standard errors does not eliminate heteroskedasticity in the data. Rather, it corrects the inferential procedure so that statistical conclusions are no longer affected by violations of the homoskedasticity assumption. Consequently, formal heteroskedasticity tests are not reapplied after the use of robust standard errors.

TABEL V. ROBUST OLS (HC3) RESULTS

Variable	Coefficient	Std. Error	t-Statistic	p-Value
Intercept	13.816	8.714	1.586	0.121
Age (Usia)	-0.384	0.377	-1.018	0.315
Sleep Hours per Day	-0.562	0.312	-1.804	0.079
Social Media Hours per Day	-0.092	0.196	-0.468	0.643
Academic Credit Load (SKS)	-0.030	0.147	-0.204	0.840
Work Hours per Day	0.255	0.158	1.613	0.115
Organization Hours per Day	0.886	0.256	3.461	0.001
Commute Minutes per Day	0.008	0.014	0.552	0.584
Exercise Hours per Day	0.019	0.240	0.080	0.937

Table V presents the estimation results of the OLS model with heteroskedasticity robust standard errors (HC3). As expected, the coefficient estimates remain identical to those reported in the baseline OLS model, since the HC3 procedure does not modify the point estimates. The primary difference lies in the estimated standard errors, which are adjusted to account for the presence of heteroskedasticity.

After applying robust standard errors, changes in statistical significance are observed for several variables. Compared to the baseline OLS results, the robust standard errors are generally larger, indicating that the conventional OLS standard errors understated the true variability of some coefficient estimates. This confirms that inference based solely on the baseline OLS model may be unreliable.

Under robust inference, organization hours per day remains statistically significant at the 1% level, indicating a stable and robust relationship with daily study hours that is not sensitive to heteroskedasticity. Other variables that appeared statistically significant in the baseline OLS model exhibit weaker significance or lose significance once robust standard errors are applied. This suggests that their apparent significance in the baseline model may be driven by underestimated standard errors rather than strong underlying relationships.

Importantly, the robust OLS results do not imply that heteroskedasticity has been eliminated. Instead, they ensure that hypothesis testing based on t-statistics and p-values remains valid despite the violation of the homoskedasticity assumption. For this reason, no further heteroskedasticity tests are conducted after the application of HC3 standard errors.

Based on these considerations, the robust OLS model is regarded as providing more reliable statistical inference than the baseline OLS model and is therefore used as the primary benchmark in subsequent model comparisons.

2. Weighted Least Squares (WLS)

a. Weighted Least Squares Estimation (Normal)

As an alternative approach to handling heteroskedasticity, the model is estimated using Weighted Least Squares (WLS). In this method, observations are weighted inversely to the squared residuals obtained from the baseline OLS model. By assigning lower weights to observations with higher error variance, WLS aims to improve estimation efficiency under heteroskedasticity.

TABEL VI. WLS ESTIMATION RESULTS

Variable	Coefficient	Std. Error	t-Statistic	p-Value
Intercept	13.5606	3.0505	4.445	0.0001
Age (Usia)	-0.3375	0.0829	-4.075	0.0002
Sleep Hours per Day	-0.6661	0.1263	-5.275	0.0000

Social Media Hours per Day	-0.1756	0.0381	-4.609	0.0000
Academic Credit Load (SKS)	-0.0206	0.0485	-0.425	0.6736
Work Hours per Day	0.2262	0.0268	8.436	0.0000
Organization Hours per Day	0.8501	0.0323	26.290	0.0000
Commute Minutes per Day	0.0048	0.0043	1.116	0.2719
Exercise Hours per Day	0.2565	0.1197	2.142	0.0391

The WLS results show substantially smaller standard errors and a very high coefficient of determination. However, such results may reflect overfitting or misspecification of the weighting scheme, particularly when weights are derived from estimated residuals. Therefore, inference based solely on conventional WLS standard errors may be overly optimistic.

b. Robust Weighted Least Squares

To mitigate potential misspecification of the WLS variance structure, the WLS model is re-estimated using heteroskedasticity robust standard errors. This approach provides consistent inference even if the weighting scheme does not fully capture the true form of heteroskedasticity.

TABEL VII. ROBUST WLS RESULTS

Variable	Coefficient	Std. Error	t-Statistic	p-Value
Intercept	13.5606	3.9926	3.396	0.0017
Age (Usia)	-0.3375	0.1056	-3.197	0.0029
Sleep Hours per Day	-0.6661	0.1465	-4.548	0.0001
Social Media Hours per Day	-0.1756	0.0426	-4.119	0.0002
Academic Credit Load (SKS)	-0.0206	0.0647	-0.319	0.7518
Work Hours per Day	0.2262	0.0458	4.940	0.0000
Organization Hours per Day	0.8501	0.0411	20.699	0.0000
Commute Minutes per Day	0.0048	0.0049	0.990	0.3288
Exercise Hours per Day	0.2565	0.1480	1.733	0.0917

After applying robust standard errors, the magnitude of standard errors increases for several variables, while coefficient estimates remain unchanged. This indicates that inference based on conventional WLS standard errors may overstate statistical significance. The robust WLS results therefore provide a more reliable basis for comparison with the robust OLS model.

3. Generalized Least Squares (GLS)

As an alternative approach to handling heteroskedasticity, the model is estimated using Generalized Least Squares (GLS). Unlike OLS and WLS, GLS explicitly accounts for potential heteroskedasticity by modeling the variance structure of the error term. This approach allows for more efficient estimation when the form of heteroskedasticity can be reasonably approximated. The GLS model is estimated using Restricted Maximum Likelihood (REML), and the estimation results are reported in Table VIII.

TABEL VIII. GLS ESTIMATION RESULTS

Variable	Coefficient	Std. Error	t-Statistic	p-Value
Intercept	13.2302	7.8230	1.691	0.0994
Age (Usia)	-0.3570	0.3481	-1.026	0.3119
Sleep Hours per Day	-0.5773	0.2524	-2.287	0.0282
Social Media Hours per Day	-0.1182	0.1280	-0.924	0.3618
Academic Credit Load (SKS)	-0.0271	0.1381	-0.196	0.8456
Work Hours per Day	0.2564	0.1023	2.506	0.0169
Organization Hours per Day	0.8816	0.1468	6.006	0.0000
Commute Minutes per Day	0.0088	0.0072	1.213	0.2331
Exercise Hours per Day	0.2550	0.4091	0.623	0.5370

The GLS estimates display coefficient signs and magnitudes that are largely consistent with those obtained from the baseline OLS, robust OLS, and robust WLS models. This consistency suggests that the estimated relationships are stable across different estimation techniques.

However, the GLS model does not yield a substantial improvement in model fit relative to the robust OLS specification, as indicated by the information criteria. Given that GLS requires stronger assumptions regarding the structure of the error variance, its use is primarily intended as a comparison model rather than the main analytical specification.

4. Model Comparison

This section compares the OLS, robust OLS, robust WLS, and GLS estimators to determine the most appropriate model for subsequent analysis.

The baseline OLS model provides an initial benchmark and demonstrates good overall explanatory power. However, diagnostic testing indicates the presence of heteroskedasticity, rendering conventional OLS standard errors unreliable for statistical inference. As a result, the baseline OLS model is not suitable for final inference.

The robust OLS model (HC3) corrects the standard errors without altering the coefficient estimates, thereby ensuring valid inference under heteroskedasticity. This approach does not impose assumptions on the functional

form of the error variance and remains stable across specifications. For this reason, robust OLS serves as the primary reference model in the comparison.

The WLS estimator yields very high explanatory power and substantially smaller standard errors. However, its results are sensitive to the specification of the weighting scheme. Once robust standard errors are applied to the WLS model, several coefficients exhibit reduced statistical significance, indicating that conventional WLS inference may overstate precision. Consequently, robust WLS is treated as a robustness check rather than a preferred estimator.

The GLS model produces coefficient estimates and significance patterns that are broadly consistent with those of the robust OLS model. However, GLS requires stronger assumptions regarding the structure of heteroskedasticity and does not provide a clear improvement in model fit relative to robust OLS. Therefore, GLS is retained as a comparison model rather than the main analytical specification.

Based on these considerations, the robust OLS model with HC3 standard errors is selected as the final estimation strategy. This model offers a balance between reliability of inference, theoretical completeness, and robustness to heteroskedasticity, making it the most appropriate choice for addressing the research questions in the subsequent analysis.

So, final model selection = robust OLS model with HC3 standard errors.

F. Discussion of All Research Questions

This section discusses the findings of the study by addressing the three research questions based on the results of the selected estimation strategy. Following the model selection process, the discussion primarily refers to the robust OLS model with HC3 standard errors, as reported in Table V, which provides reliable statistical inference in the presence of heteroskedasticity.

1. Research Question 1: How does the use of heteroskedasticity-robust regression models improve the accuracy of estimating students' daily study hours?

Discussion:

The use of heteroskedasticity-robust regression models improves estimation accuracy by ensuring valid statistical inference when the assumption of constant error variance is violated. Diagnostic testing of the baseline OLS model (Table IV) indicates the presence of heteroskedasticity, implying that the conventional OLS standard errors may be inconsistent and lead to misleading conclusions.

As shown in Table V, the application of HC3 robust standard errors does not alter the estimated coefficients but results in larger standard errors for several variables. Consequently, some predictors that appear statistically significant under baseline OLS lose significance once robust inference is applied. This indicates that the baseline OLS model may overstate the precision of certain estimates.

Therefore, heteroskedasticity-robust regression models improve accuracy not by changing point estimates, but by correcting the inferential procedure. This leads to more conservative and credible conclusions regarding the determinants of students' daily study hours.

2. Research Question 2: Which variable predictor has the strongest influence on study hours?

Discussion:

Based on the robust OLS estimation results presented in Table V, organization hours per day emerge as the strongest predictor of students' daily study hours. This variable remains statistically significant at the 1% level under robust OLS estimation and exhibits a consistently large coefficient magnitude.

The robustness of this result is further supported by the estimation outcomes from robust WLS (Table VII) and GLS (Table VIII), where organizational activity hours continue to show strong statistical significance. The consistency of this finding across different estimation methods indicates that the influence of organizational involvement on study hours is stable and not sensitive to model specification.

3. Research Question 3: How do these eight variables affect daily study hours in a multivariate regression framework?

Discussion:

In a multivariate regression framework, the eight explanatory variables jointly account for students' time allocation, academic demands, and extracurricular engagement. As indicated by the robust OLS results in Table V, these variables collectively explain a substantial proportion of the variation in daily study hours, as reflected by a relatively high adjusted R^2.

The estimated coefficients in Table V represent partial effects, meaning that each variable's association with study hours is evaluated while holding other factors constant. Within this framework, organizational activity hours show a strong and positive association with study time, while sleep duration exhibits a negative relationship. Work hours display a positive but weaker association under robust inference.

Other variables including age, social media usage, academic credit load (SKS), commuting time, and exercise duration do not exhibit statistically significant effects under robust OLS estimation. Nevertheless, as retained in the full specification, these variables serve as important controls that reduce the risk of omitted variable bias and ensure theoretical completeness of the model.

V. Conclusion

This study examines the factors influencing students' daily study hours outside of class using a multivariate regression framework. To address violations of the homoskedasticity assumption, heteroskedasticity-robust regression techniques were employed to ensure reliable statistical inference.

The results indicate that accounting for heteroskedasticity substantially improves the credibility of inference by correcting underestimated standard errors in conventional OLS estimation. Among the examined predictors, organizational activity hours consistently emerge as the strongest determinant of study time across multiple estimation methods. Sleep duration and work hours also exhibit meaningful associations, although their significance is sensitive to the choice of estimator.

Overall, the findings highlight the importance of structured activities in shaping students' study behavior and demonstrate the value of robust estimation techniques in empirical studies using survey data. Future research may extend this analysis by incorporating longitudinal data or alternative modeling approaches to further explore causal mechanisms underlying students' time allocation decisions.

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Appendix A: R Code

Necessary Libraries

```
library(readxl)
library(MASS)
library(dplyr)
library(psych)
library(nortest)
library(lmtest)
library(car)
library(sandwich)
```

Data Preparation

```
#1. Data Import and Initial Cleaning
data_raw <-
    read_excel("/Users/vaniawyanet/Documents/SE
    MS_5/regression analysis/Survei.xlsx")
head(data_raw)

# cek unique value from status
unique(data_raw$status)
table(tolower(trimws(data_raw$status)))

# Retain only respondents with student status
data_mhs <- data_raw %>%
  filter(tolower(status) == "mahasiswa")
unique(data_mhs$status)

# Remove non-analytical variables
    (identifiers)
data_clean <- data_mhs %>%
  dplyr::select(-nama, -status)

data_clean <- data_clean %>%
  mutate(across(everything(), as.numeric))

#Check missing value
colSums(is.na(data_clean))

#Boxplot for outlier screening
p <- ceiling(sqrt(ncol(data_clean)))
par(mfrow = c(p, p))

for (v in names(data_clean)) {
  boxplot(
    data_clean[[v]],
    main = paste("Boxplot of", v),
    col = "lightblue",
    horizontal = FALSE
  )
}

# Detect outlier using IQR method
detect_iqr_outlier <- function(x) {
  Q1 <- quantile(x, 0.25, na.rm = TRUE)
  Q3 <- quantile(x, 0.75, na.rm = TRUE)
  IQR <- Q3 - Q1

  which(x < (Q1 - 1.5 * IQR) | x > (Q3 + 1.5 *
    IQR))
}
out_sks <- detect_iqr_outlier(data_clean$SKS)
out_exercise <-
    detect_iqr_outlier(data_clean$ExerciseHours
    PerDay)

# Inspect detected outlier values
data_clean[out_sks, "SKS"]
data_clean[out_exercise,
    "ExerciseHoursPerDay"]


# Remove extreme but implausible outliers
data_clean <- data_clean %>%
  filter(SKS != 88, ExerciseHoursPerDay != 24)

# Verify cleaned dataset
nrow(data_clean)
str(data_clean)
head(data_clean)

# 2. Exploratory Data Analysis (EDA)
d <- density(data_clean$StudyHoursPerDay)

# Plot histogram with density overlay
hist(
  data_clean$StudyHoursPerDay,
  probability = TRUE,
  breaks = 10,
  col = rgb(0.7, 0.85, 0.9, 0.6),
  border = "white",
  main = "Distribution of Study Hours per
    Day",
  xlab = "Study Hours per Day",
  ylim = c(0, max(d$y) * 1.1)  # buffer 10%
)

# Overlay kernel density curve
lines(d, lwd = 2, col = "black")

# 3. Initial OLS Model and Influential
    Observation Detection

# Fit full multiple linear regression model
model <- lm(StudyHoursPerDay ~ ., data =
    data_clean)
summary(model)

# Detect extreme standardized residuals
    (response outliers)
std_resid <- rstandard(model)
which(abs(std_resid) > 3)

# Detect high-leverage observations
hatvalues(model)
which(hatvalues(model) > 2 * (8 + 1) / 47)

# Detect influential observations using Cook's
    distance
```

```
cook <- cooks.distance(model)
which(cook > 1)

# 4. Removal of Implausible and Influential
    Observations

# Remove observations with implausible input
    values
# Observation 5  : ExerciseHoursPerDay = 24
    (not realistic)
# Observation 47 : SKS = 88 (invalid for per-
    semester credit load)
data_clean <- data_clean[-c(5, 47), ]

# Updated sample size
nrow(data_clean)

# 5. Final OLS Model after Data Cleaning

# Refit regression model
model <- lm(StudyHoursPerDay ~ ., data =
    data_clean)
summary(model)

# Re-check outliers and influence
which(abs(rstandard(model)) > 3)
which(hatvalues(model) > 2 * (8 + 1) / 45)
which(cooks.distance(model) > 1)
```

Model Evaluation

```
# ANOVA table
anova(model)

# R-squared and Adjusted R-squared
summary(model)$r.squared
summary(model)$adj.r.squared

# 95% Confidence Intervals for regression
    coefficients
confint(model, level = 0.95)

# t-statistics and p-values
summary(model)$coefficients
```

Correlation Analysis

```
# Correlation among predictors
X <- data_clean[, c("Usia",
    "SleepHoursPerDay",
    "SocialMediaHoursPerDay",
                    "SKS", "WorkHoursPerDay",
    "OrganizationHoursPerDay",
                    "CommuteMinutesPerDay",
    "ExerciseHoursPerDay")]

corr.test(X)

# Correlation including dependent variable
corr.test(data_clean)
```

Assumption Checking

```
# Normality of residuals (Lilliefors test)
e <- resid(model)
lillie.test(e)

# Heteroskedasticity test (Breusch-Pagan)
bptest(model)

# Autocorrelation test (Durbin-Watson)
dwtest(model)

# Multicollinearity check (VIF)
vif(model)
```

Variable Selection

```
# Forward Selection

# Step 1: Fit all single-predictor models
m1 <- lm(StudyHoursPerDay ~ Usia, data =
    data_clean)
m2 <- lm(StudyHoursPerDay ~ SleepHoursPerDay,
    data = data_clean)
m3 <- lm(StudyHoursPerDay ~
    SocialMediaHoursPerDay, data = data_clean)
m4 <- lm(StudyHoursPerDay ~ SKS, data =
    data_clean)
m5 <- lm(StudyHoursPerDay ~ WorkHoursPerDay,
    data = data_clean)
m6 <- lm(StudyHoursPerDay ~
    OrganizationHoursPerDay, data = data_clean)
m7 <- lm(StudyHoursPerDay ~
    CommuteMinutesPerDay, data = data_clean)
m8 <- lm(StudyHoursPerDay ~
    ExerciseHoursPerDay, data = data_clean)

# Review outputs (used to identify the most
    significant single predictor)
summary(m1)
summary(m2)
summary(m3)
summary(m4)
summary(m5)
summary(m6)
summary(m7)
summary(m8)

# Selected predictor for Step 2:
# OrganizationHoursPerDay (smallest p-value
    among single-predictor models)

# Step 2: Fit two-predictor candidate models
# Base predictor: OrganizationHoursPerDay
m61 <- lm(StudyHoursPerDay ~
    OrganizationHoursPerDay + Usia, data =
    data_clean)
m62 <- lm(StudyHoursPerDay ~
    OrganizationHoursPerDay + SleepHoursPerDay,
    data = data_clean)
m63 <- lm(StudyHoursPerDay ~
    OrganizationHoursPerDay +
    SocialMediaHoursPerDay, data = data_clean)
m64 <- lm(StudyHoursPerDay ~
    OrganizationHoursPerDay + SKS, data =
    data_clean)
m65 <- lm(StudyHoursPerDay ~
    OrganizationHoursPerDay + WorkHoursPerDay,
    data = data_clean)
m67 <- lm(StudyHoursPerDay ~
    OrganizationHoursPerDay +
    CommuteMinutesPerDay, data = data_clean)
m68 <- lm(StudyHoursPerDay ~
    OrganizationHoursPerDay +
    ExerciseHoursPerDay, data = data_clean)
```

```
# Review outputs (used to identify the best
    second predictor)
summary(m61)
summary(m62)
summary(m63)
summary(m64)
summary(m65)
summary(m67)
summary(m68)

# Selected predictor for Step 3:
# SleepHoursPerDay (model m62 shows
    statistically significant contribution)

# Step 3: Fit three-predictor candidate models
# Base predictors: OrganizationHoursPerDay +
    SleepHoursPerDay
m621 <- lm(StudyHoursPerDay ~
    OrganizationHoursPerDay + SleepHoursPerDay
    + Usia,
           data = data_clean)
m623 <- lm(StudyHoursPerDay ~
    OrganizationHoursPerDay + SleepHoursPerDay
    + SocialMediaHoursPerDay,
           data = data_clean)
m624 <- lm(StudyHoursPerDay ~
    OrganizationHoursPerDay + SleepHoursPerDay
    + SKS,
           data = data_clean)
m625 <- lm(StudyHoursPerDay ~
    OrganizationHoursPerDay + SleepHoursPerDay
    + WorkHoursPerDay,
           data = data_clean)
m627 <- lm(StudyHoursPerDay ~
    OrganizationHoursPerDay + SleepHoursPerDay
    + CommuteMinutesPerDay,
           data = data_clean)
m628 <- lm(StudyHoursPerDay ~
    OrganizationHoursPerDay + SleepHoursPerDay
    + ExerciseHoursPerDay,
           data = data_clean)

# Review outputs (used to check whether adding
    a third predictor improves the model)
summary(m621)
summary(m623)
summary(m624)
summary(m625)
summary(m627)
summary(m628)

# Decision:
# Since none of the additional third
    predictors is statistically significant at
    alpha = 0.05,
# the final forward-selected model remains the
    two-predictor specification (m62).

# Final Forward-Selected Model
summary(m62)


####################
# Backward Elimination

# Step 0: Full model (8 predictors)
m12345678 <- lm(StudyHoursPerDay ~ .,  data =
    data_clean)
summary(m12345678)

# Based on the full model summary, remove the
    predictor with the largest p-value:
# Removed: SKS

# Step 1: 7 predictors (remove SKS)
m1235678 <- lm(
  StudyHoursPerDay ~ Usia + SleepHoursPerDay +
    SocialMediaHoursPerDay +
    WorkHoursPerDay + OrganizationHoursPerDay
    + CommuteMinutesPerDay +
    ExerciseHoursPerDay,
  data = data_clean
)
summary(m1235678)

# Removed: ExerciseHoursPerDay (largest p-
    value)

# Step 2: 6 predictors (remove
    ExerciseHoursPerDay)
m123567 <- lm(
  StudyHoursPerDay ~ Usia + SleepHoursPerDay +
    SocialMediaHoursPerDay +
    WorkHoursPerDay + OrganizationHoursPerDay
    + CommuteMinutesPerDay,
  data = data_clean
)
summary(m123567)

# Removed: SocialMediaHoursPerDay (largest p-
    value)

# Step 3: 5 predictors (remove
    SocialMediaHoursPerDay)
m12567 <- lm(
  StudyHoursPerDay ~ Usia + SleepHoursPerDay +
    WorkHoursPerDay +
    OrganizationHoursPerDay +
    CommuteMinutesPerDay,
  data = data_clean
)
summary(m12567)

# Removed: CommuteMinutesPerDay (largest p-
    value)

# Step 4: 4 predictors (remove
    CommuteMinutesPerDay)
m1256 <- lm(
  StudyHoursPerDay ~ Usia + SleepHoursPerDay +
    WorkHoursPerDay +
    OrganizationHoursPerDay,
  data = data_clean
)
summary(m1256)

# Removed: Usia (largest p-value)

# Step 5: 3 predictors (remove Usia)
m256 <- lm(
  StudyHoursPerDay ~ SleepHoursPerDay +
    WorkHoursPerDay +
    OrganizationHoursPerDay,
```

```
  data = data_clean
)
summary(m256)

# Removed: WorkHoursPerDay (largest p-value;
    marginal > 0.05)

# Step 6: Final reduced model (2 predictors)
m26 <- lm(
  StudyHoursPerDay ~ SleepHoursPerDay +
    OrganizationHoursPerDay,
  data = data_clean
)
summary(m26)

# Optional: Display the selected reduced model
    vs. full model
summary(m26)
summary(m12345678)
```

Model Selection

```
model_ols <- lm(
  StudyHoursPerDay ~ Usia +
    SleepHoursPerDay +
    SocialMediaHoursPerDay +
    SKS +
    WorkHoursPerDay +
    OrganizationHoursPerDay +
    CommuteMinutesPerDay +
    ExerciseHoursPerDay,
  data = data_clean
)

summary(model_ols)

# Tes heteroskedastisitas
library(lmtest)
bptest(model_ols)

# Robust Standard Errors (HC3)
model_robustSE <- coeftest(model_ols, vcov =
    vcovHC(model_ols, type = "HC3"))
model_robustSE

# MODEL 1: OLS
res_ols <- resid(model_ols)
epsilon <- 1e-6
w_wls <- 1 / (res_ols^2 + epsilon)

# MODEL 2: WLS
model_wls <- lm(
  StudyHoursPerDay ~ Usia +
    SleepHoursPerDay +
    SocialMediaHoursPerDay +
    SKS +
    WorkHoursPerDay +
    OrganizationHoursPerDay +
    CommuteMinutesPerDay +
    ExerciseHoursPerDay,
  data = data_clean,
  weights = w_wls
)

summary(model_wls)
bptest(model_wls)

# Robust WLS (RWLS)
model_rwls <- coeftest(model_wls, vcov =
    vcovHC(model_wls, type = "HC3"))
model_rwls

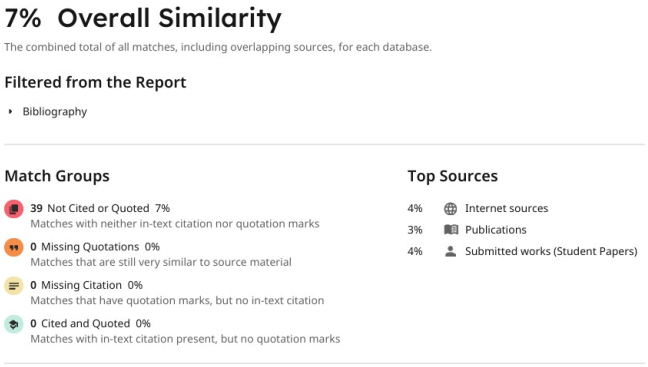
# MODEL 3: Generalized Least Squares (GLS)
library(nlme)

model_gls <- gls(
  StudyHoursPerDay ~ Usia +
    SleepHoursPerDay +
    SocialMediaHoursPerDay +
    SKS +
    WorkHoursPerDay +
    OrganizationHoursPerDay +
    CommuteMinutesPerDay +
    ExerciseHoursPerDay,
  data = data_clean,
  weights = varIdent(form = ~ 1)
)

summary(model_gls)

plot(fitted(model_gls), resid(model_gls))
abline(h = 0, col="red")
```

Appendix B : Turnitin Results



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